

Building a Docker image

As we now have an application and Dockerfile, we are ready to build a Docker image for the sample application. Make sure you are in the root directory of your application before building an image using the following command:

```
docker build -t node-docker
```

```
C:\ProgramData\Docker\node-docker>docker build -t node-docker .
Sending build context to Docker daemon 6.935MB
Step 1/7 : FROM node:10
--> 4c447b45159a
Step 2/7 : WORKDIR /usr/src/app
--> Using cache
--> a26744fa3e31
Step 3/7 : COPY package*.json ./
--> Using cache
--> 61d89ff552e4
Step 4/7 : RUN npm install
--> Using cache
--> 0a0f4a80679f
Step 5/7 : COPY . .
--> 699c0970c85f
Step 6/7 : EXPOSE 8080
--> Running in 8e3aab9c7674
Removing intermediate container 8e3aab9c7674
--> cc5d888cd8b3
Step 7/7 : CMD [ "npm", "start" ]
--> Running in 4f317a32c234
Removing intermediate container 4f317a32c234
--> b7537396683d
Successfully built b7537396683d
Successfully tagged node-docker:latest
```

Figure 1: Building a Docker image

You will now be able to see the execution of each step mentioned in the Dockerfile after successfully building the image that you will see below the message on your command line, as shown in Figure 1.

Running containers

After we have successfully built the container, we need to run it. Execute the following command to run the container. We will also need to map the port for the container.

```
docker run -p 33221:3000 -d node-docker
```

```
C:\ProgramData\Docker\node-docker>docker ps
CONTAINER ID        IMAGE               COMMAND
9bcb105a5f3fd82217a86c869d04fe57
```

Figure 2: Running the container

With the `-d` option, we are instructing the container to run in the background and print the container ID; and `-p` is used to publish a container's port to the host.

After we are satisfied with how the container works, we can stop the

container using the string given below:

```
docker stop <CONTAINER_ID>
```

The container ID can be found by the command that lists containers:

```
docker ps
```

```
C:\ProgramData\Docker\node-docker>docker ps
CONTAINER ID        IMAGE               COMMAND
9bcb105a5f3fd82217a86c869d04fe57
```

Figure 3: Listing containers

Sharing containers

Now that our container is ready for sharing, we can do so by using any of the available Docker registries like Docker Hub or GitHub. These are the few popular ones on which we can host public or private Docker images. We will use Docker Hub here, but before pushing an image to it, make sure you have already signed up for an account, since Docker Hub credentials are required to build the image. Now execute the following command to build the image using Docker Hub.

```
docker build -t[Username]/node-docker
```

Log in to Docker Hub using the command shown below:

```
docker login
```

You will be asked for your credentials if you are logging in for the first time, else your existing credentials will be used. Now we will push the image to Docker Hub using the command below:

```
docker push [username]/node-docker
```

After the image has been successfully pushed, visit the Docker Hub website at <https://hub.docker.com>, where you can see the pushed

images in the repositories. Now this image can be used on any server with Docker installed, just by using the `run` command.

To pull our Docker image `username/node-docker`, execute the following command:

```
docker pull [username]/node-docker
```

Now we will just need to run our container from the image.

```
docker run -p 31221:3000 -d [username]/node-docker
```

```
C:\ProgramData\Docker\node-docker>docker run -p 31221:3000 -d [username]/node-docker
16c266940a6b40b15adb3c592d9bcfb1db862fa1
```

Figure 4: Running a Docker image from the shared container

You will be able to browse this application at <http://127.0.0.1:31221/>.

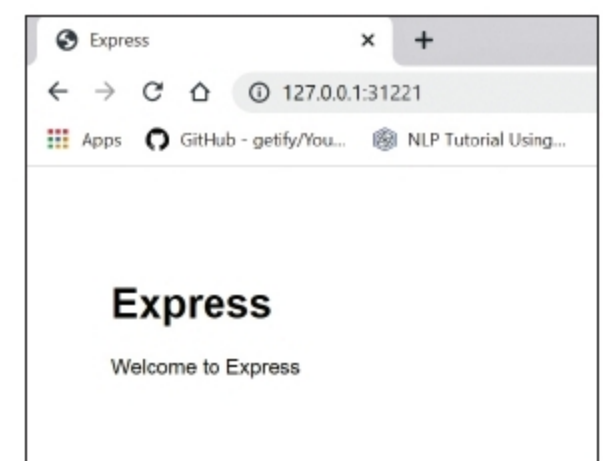


Figure 5: Dockerised application

That's all. You have now successfully Dockerised a Node.js application. **END** 🐧

References

- [1] <https://docs.docker.com>
- [2] <https://nodejs.org>
- [3] <https://nodesource.com/blog/containerizing-node-js-applications-with-docke>

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